

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY (ICDFA)

Intro to Critical Infrastructure for IT Learners

LAB 05

Building a SCADA Human Machine Interface

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1. Objective

The objective of this laboratory was to build a basic SCADA Human Machine Interface (HMI) in FUXA and connect it to an OpenPLC tank-fill simulation.

The HMI was configured to display the tank level and valve state and to reflect changes from the running PLC process.

2. Implementation

The HMI was created in FUXA using a tank, tank-level bar gauge, motor-actuated valve and process piping.

The OpenPLC tank-fill simulation was started and a Modbus TCP connection was configured between OpenPLC and FUXA.

The following tags were configured:

- **Tank Level** – Int16 value representing the tank level.
- **Tank Valve Open** – Boolean value representing the valve open/closed state.

The tank-level gauge was configured with a range of 0–10 ft and was bound to the **Tank Level** tag.

The valve graphic was bound to the **Tank Valve Open** tag. The HMI was configured so that the valve has a closed representation when the value is 0 and an open representation when the value is 1.

3. Evidence

3.1 OpenPLC/FUXA Connection and Tags

The following screenshot shows the configured OpenPLC/FUXA connection and the required PLC tags.

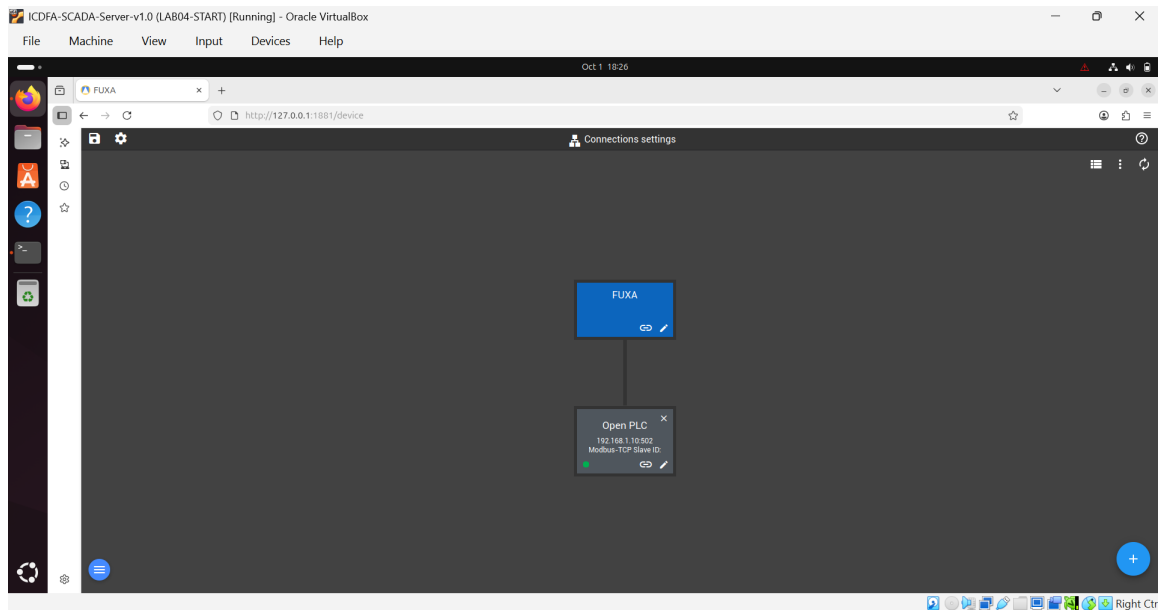


Figure 1: Configured OpenPLC/FUXA connection and PLC tags.

3.2 FUXA Design View

The following screenshot shows the completed HMI design in FUXA edit mode.

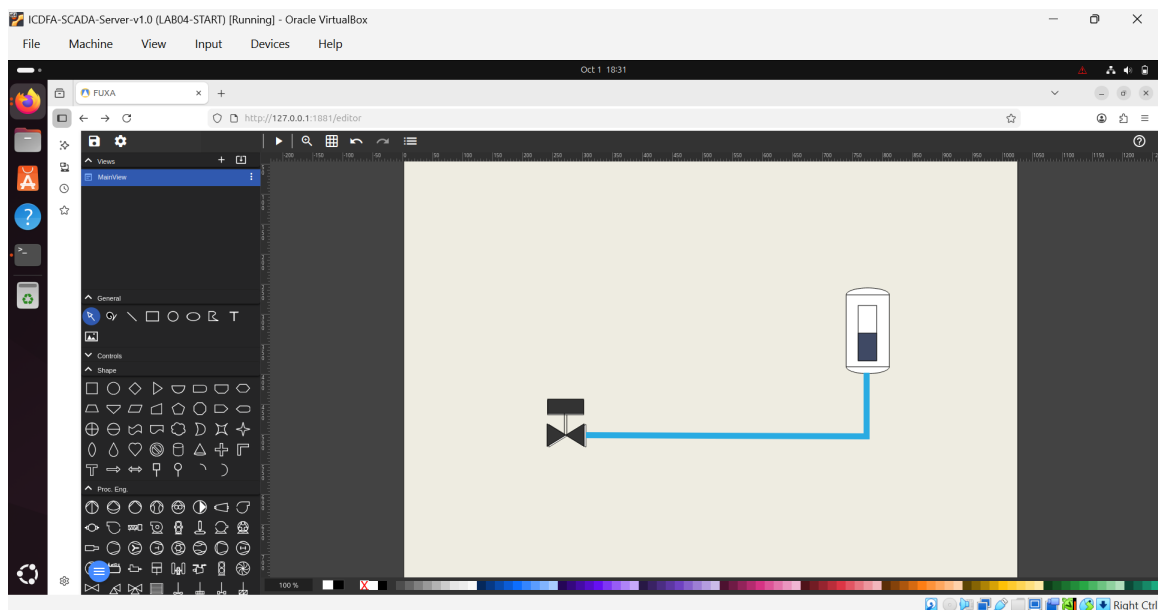


Figure 2: Completed FUXA HMI design view.

3.3 Running HMI – Evidence 1

The following screenshot shows the HMI running with the tank-level visualization.

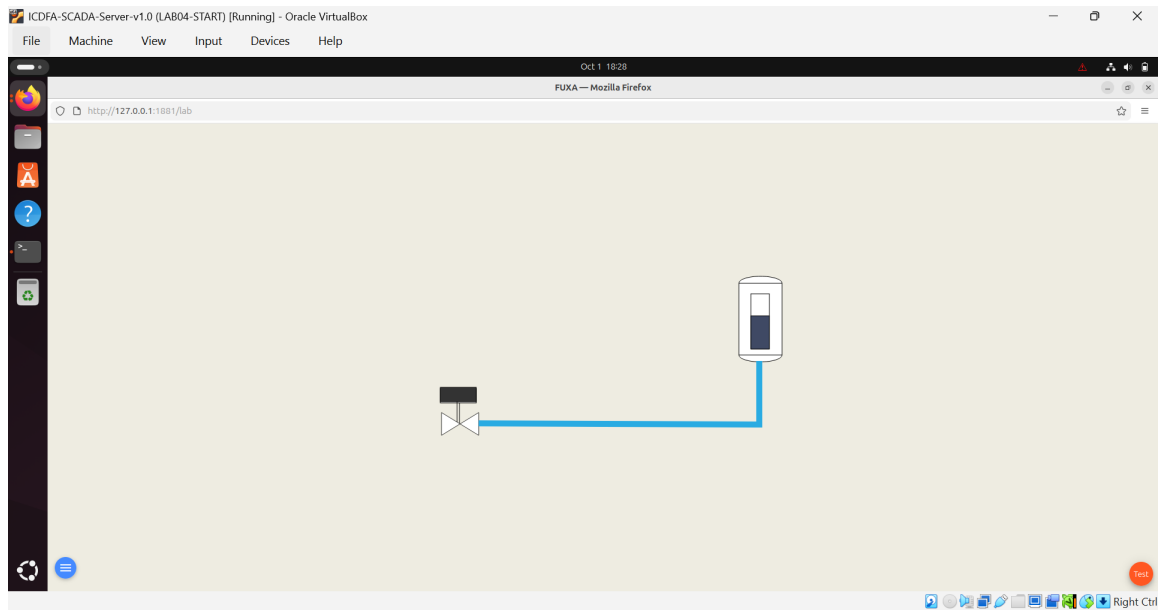


Figure 3: Running HMI showing the tank and level indication.

3.4 Running HMI – Evidence 2

The following screenshot provides another operating state of the HMI.

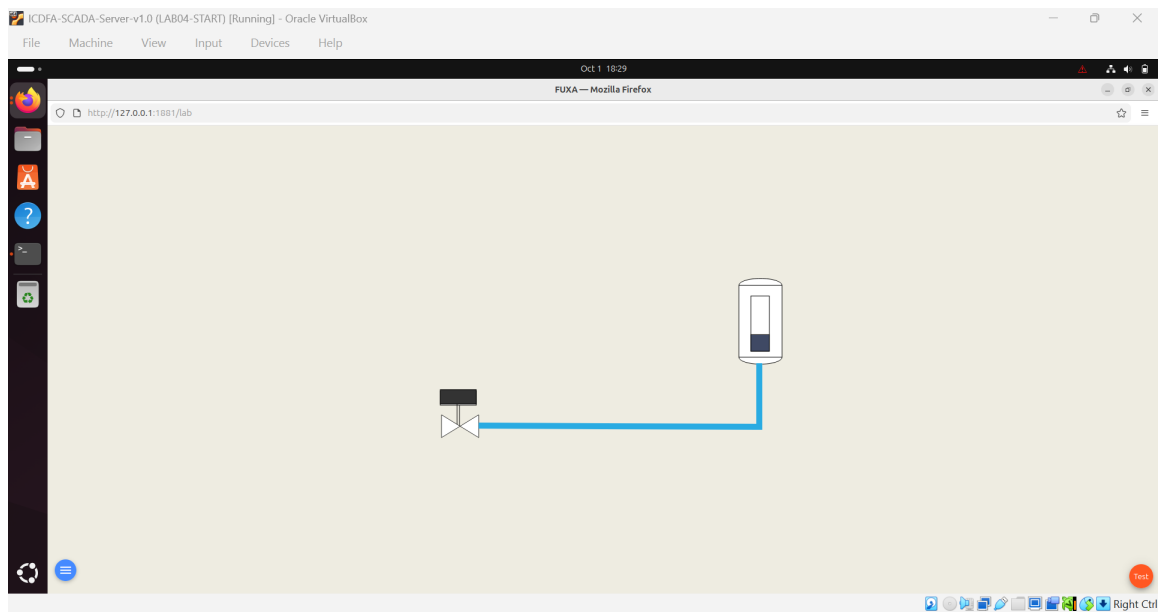


Figure 4: Running HMI showing the valve/process state.

3.5 Running HMI – Evidence 3

The following screenshot provides additional evidence that the HMI reflects the running process.

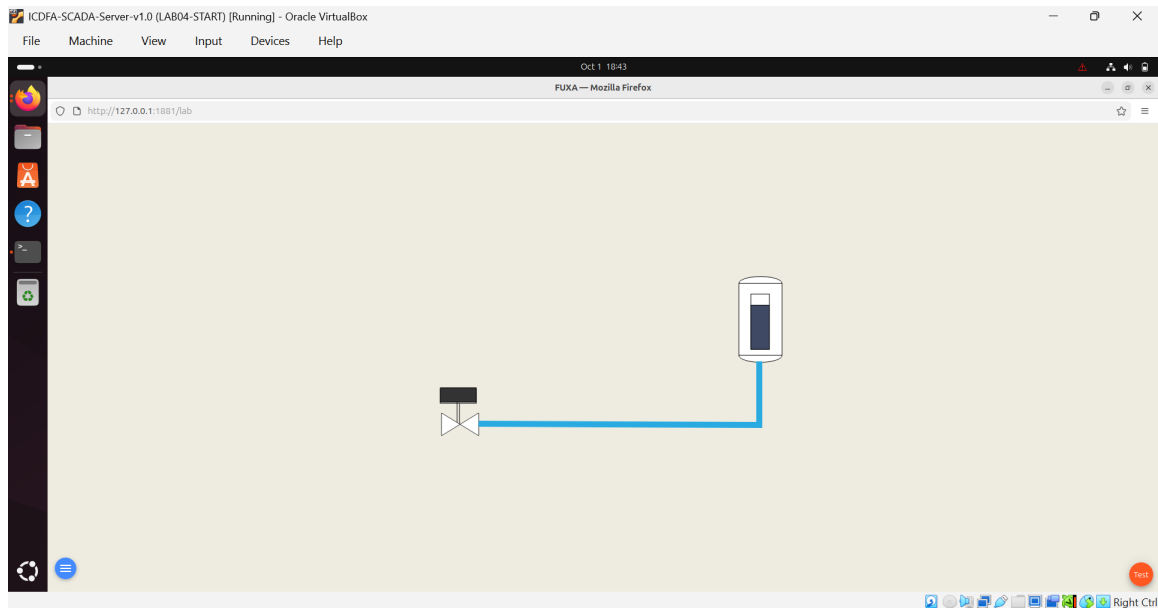


Figure 5: Running HMI showing the process state during operation.

4. Design Rationale

The HMI was kept simple and process-oriented. The tank is the main process object, with the level indication placed inside the tank so that the operator can easily relate the displayed level to the process.

The valve and pipe were positioned to show the direction of the tank-filling process. A neutral visual style was used for the normal process state, while the valve changes appearance according to its open or closed condition.

This design helps the operator quickly understand the tank level and valve state without unnecessary animation or excessive use of color.

5. Knowledge Check

1. What is a P&ID and how does it influence HMI layout?

A P&ID (Piping and Instrumentation Diagram) is a diagram that represents process equipment, piping, instrumentation and control components in an industrial process.

It influences HMI layout by providing a reference for the arrangement and relationship of the process components. In this laboratory, the tank, valve, pipe and level indication were arranged to represent the actual tank-filling process.

2. Why do high-performance HMIs avoid unnecessary animation and excessive color?

Unnecessary animation and excessive color can distract the operator and make important process information harder to recognize.

A simple and restrained interface allows meaningful changes, alarms and abnormal conditions to receive attention when necessary.

3. How does the Tank Level tag differ from the Tank Valve Open tag in data type and behavior?

The `Tank Level` tag is an `Int16` numerical value representing the tank level. It can change across a range of values and is used by the tank-level gauge.

The `Tank Valve Open` tag is a Boolean value with two states:

- 0 – valve closed
- 1 – valve open

Therefore, Tank Level represents a changing numerical measurement, while Tank Valve Open represents a discrete state.

4. What could happen operationally if an HMI displays stale or incorrect PLC data?

Stale or incorrect PLC data can cause the operator to have an inaccurate understanding of the actual process.

For example, the HMI could show an incorrect tank level or valve state. This could lead to inappropriate operational decisions and reduce situational awareness.

6. Conclusion

The Lab 05 SCADA HMI was successfully developed using FUXA and OpenPLC. The HMI displayed the tank level and valve state through PLC tag bindings, and the running interface reflected the simulated tank-fill process.

The laboratory provided practical experience with SCADA visualization, Modbus TCP communication, PLC tags, HMI object binding and basic high-performance HMI principles.